Math 126 C - Autumn 2021
Midterm Exam Number Two
November 18, 2021

Please try to write your name as it appears on the Canvas roster.
This makes it easier to import scans to Gradescope.

Name: _____ Student ID no. : _____

Signature: _____

1	7	
2	8	
3	15	
4	15	
5	8	
6	7	
Total	60	

This grid is purely decorative.
The exam is graded online.

- This exam consists of **SIX** problems on **FOUR** double-sided pages. The fourth page is left blank for scratch work.
- Show all work for full credit.
- You may use a TI-30X IIS (or equivalent) calculator during this exam. Other calculators and electronic devices are not permitted.
- You do not need to simplify your answers.
- If you use a trial-and-error or guess-and-check method when a more rigorous method is available, you will not receive full credit.
- Draw a box around your final answer to each problem.
- **Do not write within 1 centimeter of the edge!** Your exam will be scanned for grading.
- If you run out of room, write on one of the scratch work pages **and indicate that you have done so**. If you still need more room, raise your hand and ask for an extra page.
- You may use one hand-written double-sided 8.5″ by 11″ page of notes.
- You have 50 minutes to complete the exam.

You may use this page for scratch-work.

All work on this page will be ignored unless you write & circle "see first page" below a problem.

1. **[7 points]** A particle moves with position vector $\mathbf{r}(t) = \langle 8t + 1, t^2 - 4t, \frac{1}{2}t^3 \rangle$.
   Find the tangential and normal components of acceleration of the particle at time $t = 2$.

1 13.4

2. **[8 points]** Let $\mathcal{S}$ be the surface

$$xz + \sqrt{z} - 2xy = 6.$$

   Find the equation of the plane tangent to $\mathcal{S}$ at the point $(2, 1, 4)$.

2 14.4

3. **[15 points]** Consider the function $f(x, y) = xy^2 - 2xy + x^2$.

   Find all critical points of f, **and classify them** as local minima, local maxima, or saddle points.

3 14.7

4. **[15 points]** Let f be the function $f(x, y) = xy^2 + x^2$, and let $\mathcal{D}$ be closed disc of radius 1 centered at the origin. 4 14.7

   Find the absolute minimum and maximum values of f on $\mathcal{D}$.

5. **[8 points]** Evaluate $\int_0^5 \int_3^4 (2xy^2 + e^y)\, dy\, dx$.

NO LONGER COVERED IN MATH 126
Chapter 15 material - covered in Math 224

6. **[7 points]** Rewrite the following integral after reversing the order of integration.

   Do not try to evaluate the integral! Just reverse the order of integration.

$$\int_2^3 \int_{\frac{12}{x}}^{10-2x} \sin(y^2)\, dy\, dx$$

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